

PC-BASED WIRELESS STEPPER MOTOR CONTROL



■ V. MARIYAPPAN

Stepper motors find lots of applications in process control, machine tools and robotics. Especially in robotics and process control, it is necessary to control the stepper motor from a remote place.

Here we describe how to wirelessly control a stepper motor from a remote place by using RF modules. For this wireless stepper-motor control system, you need to design and develop the required hardware and software. The parallel port of the PC is used to control the direction of the stepper motor at the transmitter side. RF interface is used instead of IR to overcome all the drawbacks of the IR interface. The PC

signals are transmitted from the RF transmitter and received by the RF receiver.

Circuit description

Fig. 1 shows the block diagram for PC-based wireless control of a stepper motor. The signals from the parallel port of the PC are interfaced to the RF transmitter through an encoder. The encoder continuously reads the status of the relay switches, passes the data to the RF transmitter and the transmitter transmits the data. At the receiving end, the RF receiver receives this data and gives it to the decoder. The decoder converts the single-bit data into four-bit data and presents to the stepper-motor driver. Now, the driver performs the corresponding action, i.e., it rotates the stepper motor clockwise or anticlockwise.

Remote control.

For remote control, we have used the Holtek encoder-decoder pair of HT12E and HT12D. Both of these are 18-pin DIP ICs.

Operation of Holtek

HT12E and HT12D. HT12E and HT12D are CMOS ICs with a working voltage range of 2.4V to 12V.

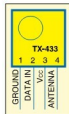


Fig. 2: Pin configuration of the RF transmitter module

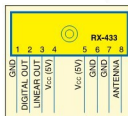


Fig. 3: Pin configuration of the RF receiver module

PARTS LIST

Semiconductors:

IC1	- HT12E Holtek encoder
IC2	- 7806, 6V regulator
IC3	- HT12D Holtek decoder
IC4	- CD40106 hex inverter
IC5	- ULN2003 Darlington array
T1-T4	- BC547 npn transistor
T5	- BC548 npn transistor
D1-D4	- 1N4148 switching diode
D5-D10	- 1N4007 rectifier diode
TX1	- TX-433 RF transmitter
RX1	- RX-433 RF receiver
LED1	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 1-mega-ohm
R2-R5	- 10-kilo-ohm
R6-R9	- 1.2-kilo-ohm
R10	- 1-kilo-ohm
R11	- 47-kilo-ohm
R12	- 3.9-kilo-ohm
R13	- 470-ohm

Capacitors:

C1, C3	- 100 μ F, 16V electrolytic
C2	- 0.1 μ F ceramic disk
C4	- 10 μ F, 16V electrolytic

Miscellaneous:

BATT.1	- 9V battery
BATT.2	- 6V, 4.5Ah battery
S1, S2	- On/off switch
DIP-SW1	- 8-way DIP switch
DIP-SW2	- 25-pin D-type male connector

TABLE I
Output Power and Current
Drain w.r.t. V_{CC}

V_{CC}	O/P	Current
5V DC	-0 dBm	1.0 mA
12V DC	+9 dBm	3 mA

TABLE II
Technical Specifications of
RF Receiver RX-433

Description	Value
Working voltage	4.5V-5.5V
Bandwidth	12 MHz
Sensitivity	-103 dBm
Data rate	4800 bps
Max. data rate	9600 bps
Standby current	1, 2 mA
Antenna	Whip, strip line or helical

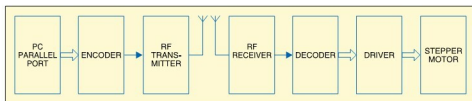


Fig. 1: Block diagram for PC-based wireless control of a stepper motor

Encoder HT12E has eight address and another four address/data lines. The data set on these twelve lines (address and address/data lines) is serially transmitted when transmit-enable pin TE is taken low. The data output appears serially on D_{OUT} pin. It is transmitted four times in succession.

The data consists of differing lengths of positive-going pulses for '1' and '0', the pulse width for '0' being twice the width of the pulse for '1'. The frequency of these pulses may lie between 1.5 and 7 kHz depending on the resistance value between OSC1 and OSC2 pins. The internal oscillator frequency of decoder HT12D is 50 times the oscillator frequency of